Integrating by parts in the second term we have

$$\begin{aligned}\beta^{-1}\int_{\mathcal{Q}} f\Delta g\bar{\rho}_\beta \mathrm{d}\omega_q &= -\beta^{-1}\int_{\mathcal{Q}} \nabla g^T\nabla(f\bar{\rho}_\beta)\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\mathcal{Q}} \nabla g^T[\nabla f\bar{\rho}_\beta - \beta\nabla U f\bar{\rho}_\beta]\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\mathcal{Q}} \nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q + \int_{\mathcal{Q}} f(\nabla g^T\nabla U)\bar{\rho}_\beta \mathrm{d}\omega_q.\end{aligned}$$

Combining this with the first part of the inner product, we see that

$$\langle f, \mathcal{L}g\rangle_\beta = -\beta^{-1}\int_{\mathcal{Q}} \nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q.$$

In this form we see that f and g appear symmetrically, so we may write

$$\langle f, \mathcal{L}g\rangle_\beta = \langle \mathcal{L}f, g\rangle_\beta,$$

hence, with respect to the weighted inner product, $\mathcal{L}^\dagger = \mathcal{L}$. (Note that the adjoint indicated here is quite different than the L^2 adjoint mentioned previously in connection with the Fokker-Planck operator.)

In addition to being self-adjoint, this operator (on the function space $C_0^2(\bar{\mu}_\beta)$) has other important properties. In particular, it has a *compact resolvent* [360]; this can be shown to imply that the spectrum consists only of a countable collection of eigenvalues, i.e. that the operator has a discrete spectrum. Moreover, from the calculation above,

$$\langle f, \mathcal{L}f\rangle = -\beta^{-1}\int_{\mathcal{Q}} \|\nabla f\|^2\bar{\rho}_\beta \mathrm{d}\omega_q,$$

so that $\langle f, \mathcal{L}f\rangle \leq 0$ so the operator is negative definite and can only have non-positive eigenvalues. Many additional results are available for this case. A good summary of the state of the art is available in the forthcoming work of G. Pavliotis [298]. In particular, it is possible to show the properties of $\mathcal{L}$ contained in Theorem 6.1.

Theorem 6.1 *Define the generator of Brownian dynamics by*

$$\mathcal{L}f \stackrel{\text{def}}{=} -\boldsymbol{M}^{-1}\nabla U^T\nabla f + \beta^{-1}\Delta_M f.$$